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Organization of the
United Nations



ICARDA
Science for resilient livelihoods in dry areas



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WEBINARS SERIES ON RAINWATER HARVESTING

19th July - 26th July - 2nd August 2022

Module n°4:

**Main guidelines for Planning
and Designing Rainwater
Harvesting systems**



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Module 4: Main guidelines for Planning and Designing Rainwater Harvesting systems

Session 2: Planning and designing of rainwater harvesting systems and structures

Date: 26th July 2022 **Time:** 13 h 00 – 14 h 30 (GMT+2)

Register: [Planning and Designing of Rainwater Harvesting Systems and Structures](#)

Session 1: Technical, socio-economic and environmental considerations in planning and designing rainwater harvesting systems

Date: 19th July 2022 **Time:** 13 h 00 – 14 h 30 (GMT+2)

Register: [Technical and Socio-Economic Considerations in Planning and Designing Rainwater Harvesting Systems](#)

Session 3: Planning and Designing of rainwater harvesting systems

Date: 2nd August 2022 **Time:** 13 h 00 – 14 h 30 (GMT+2)

Register: [Planning and Design of Rainwater Harvesting Systems](#)

BACKGROUND

Water scarcity is a critical challenge in countries and regions with arid and semi-arid conditions and fast-growing water demands. This is mostly the case in all of the countries across the NENA Region, where food demand and food insecurity continue to grow, widening the water supply-demand gap in most countries. Furthermore, the anticipated impact of climate change in the NENA region is indicative towards an increased intensity of rainstorms and flash floods, while the total amount of annual precipitation rates shows declining trends. During the last 20 years, interest has been renewed in RWH as one of the most effective climate adaptation strategies to cope with water scarcity, thus reducing the pressure on the limited freshwater resources and increasing rainfed crop productivity.

This webinar series on Rainwater Harvesting aims to provide an overview of RWH applications and illustrates, with country examples, the diversity of approaches. It will try to unlock the potential of rainwater harvesting applications in the NENA region by highlighting the role of rainwater management in overcoming the current challenges in achieving water security. It will also shed the light on the need for innovations in economics, social consensus and supportive laws and regulations to ensure a widespread uptake of RWH technologies in the NENA region.

Main Conveners:

FAO/WEPS-NENA project & Water Scarcity Initiative (WSI) with: ICARDA, ICID and FAO inter-Regional Technical Platform on Water Scarcity (iRTP-WS)

Contributing Partners:

UNESCWA and University of Florence, Italy

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Languages:

English and French with live translation

Recording:

<https://dgroups.org/fao/waterproductivity/rwh>

ABOUT THE MODULE 4

There are three webinars available for discussing the guidelines for planning and designing RWH systems. The aim is to integrate technical, engineering, socio-economic and environmental considerations in the planning and design of rainwater harvesting projects. The three webinars will discuss the key planning and design considerations through research works and project case studies in the NENA region and beyond.



EXPECTED OUTCOMES FOR THE PARTICIPANTS

Participants will achieve the following outcomes:

- 1 Recognise the importance of participatory approaches in the planning and design of rainwater harvesting;
- 2 Understand the benefits of integrated approaches in the planning and design of rainwater harvesting systems;
- 3 Comprehend the key planning and design considerations for developing sustainable rainwater harvesting systems

SUGGESTED READINGS RELATED TO SESSION 1

- **ESCWA technical booklet on rainwater harvesting:**
<https://www.unescwa.org/publications/technical-booklet-rainwater-harvesting>
- **Integrated Vulnerability Assessment Application on the Lebanese Agricultural Sector:**
<https://riccar.org/publications/integrated-vulnerability-assessment-application-lebanese-agricultural-sector>
- **Training Manual on the Integrated Vulnerability Assessment Methodology:**
<https://riccar.org/index.php/publications/training-manual-integrated-vulnerability-assessment-methodology>
- **Training Workshop on Agricultural Green Technologies in Lebanon:**
<https://www.unescwa.org/events/training-workshop-agricultural-green-technologies>
- **Promoting the use of green technologies for a resilient and a sustainable agricultural sector – Jordan:**
<https://www.unescwa.org/events/promoting-use-green-technologies-resilient-and-sustainable-agricultural-sector-jordan>
- **Promoting the use of green technologies for a resilient and a sustainable agricultural sector – Palestine:**
<https://www.unescwa.org/events/promoting-use-green-technologies-resilient-and-sustainable-agricultural-sector-palestine>
- **Laurita, Beatrice & Castelli, Giulio & Giacomini, Antonio & Salazar, Marlon & Torres, Alonso & Giambastiani, Yamuna & Preti, Federico & Bresci, Elena. (2018).** Participatory design of rooftop water harvesting systems for smallholder farmers in the corridor Seco region, Guatemala.
https://www.researchgate.net/publication/335928322_Participatory_design_of_rooftop_water_harvesting_systems_for_smallholder_farmers_in_the_corridor_Seco_region_Guatemala
- **Castelli, Giulio & Bresci, Elena. (2017).** Participatory Rural Appraisal for Diagnostic Analysis of spate irrigation systems in Raya Valley, Ethiopia. *Journal of Agriculture and Rural Development in the Tropics and Subtropics*. 118. 129-139.
https://www.researchgate.net/publication/316880179_Participatory_Rural_Appraisal_for_Diagnostic_Analysis_of_spate_irrigation_systems_in_Raya_Valley_Ethiopia
- **G. Castelli, E. Bresci, F. Castelli, E.Y. Hagos, A. Mehari (2018).** A participatory design approach for modernization of spate irrigation systems. *Agric. Water Manag.*, 210 (2018), pp. 286-295, 10.1016/j.agwat.2018.08.030
<https://www.sciencedirect.com/science/article/abs/pii/S0378377418312721#preview-section-cited-by>



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SUGGESTED READINGS RELATED TO SESSION 2

- **Oweis, Theib & Prinz, D. & Hachum, A.** (2012). Rainwater Harvesting for Agriculture in the Dry Areas. 10.1201/b12351. Chapter 6 (pp. 105-136)
https://www.researchgate.net/publication/266327820_Rainwater_Harvesting_for_Agriculture_in_the_Dry_Areas
- **Piemontese, Luigi & Castelli, Giulio & Fetzer, Ingo & Barron, Jennie & Liniger, Hanspeter & Harari, Nicole & Bresci, Elena & Jaramillo, Fernando.** (2020). Estimating the global potential of water harvesting from successful case studies. Global Environmental Change. 63. 102121. 10.1016/j.gloenvcha.2020.102121.
https://www.researchgate.net/publication/342757516_Estimating_the_global_potential_of_water_harvesting_from_successful_case_studies
- **Piemontese, Luigi & Limones, Natalia & Castelli, Giulio & Grazio, Alice & Bresci, Elena.** (2022). Large-scale siting of sand dams: a participatory approach and application in Angolan drylands. 10.22541/au.164486569.97421466/v1.
https://www.researchgate.net/publication/358601452_Large-scale_siting_of_sand_dams_a_participatory_approach_and_application_in_Angolan_drylands

SUGGESTED READINGS RELATED TO SESSION 3

- **Oweis, Theib. (2016).** Rainwater harvesting for restoring degraded dry agro-pastoral ecosystems: A conceptual review of opportunities and constraints in a changing climate. Environmental Reviews. 25. 10.1139/er-2016-0069.
https://www.researchgate.net/publication/309472461_Rainwater_harvesting_for_restoring_degraded_dry_agro-pastoral_ecosystems_A_conceptual_review_of_opportunities_and_constraints_in_a_changing_climate
- **Karrou, M., Oweis, T., Ziadat, F. and Awawdeh, F. (eds) 2011.** Rehabilitation and integrated management of dry rangelands environments with water harvesting. Community-based optimization of the management of scarce water resources in agriculture in Central and West Asia and North Africa Report no. 9. ICARDA, Aleppo, Syria vi + 208 pp.

CERTIFICATE OF ACHIEVEMENT

A digital certificate of achievement will be offered by FAO to participants who:

- Attend at least four webinars belonging to Module 1 to Module 4 (dates of the webinars: 31 May, 7, 21, and 27 June; 12, 19 and 26 July and 2 August at 01:00 p.m. GMT+2) and five sessions belonging to Module 5 to Module 10 (dates to be detailed between August and November).
- Obtain a score of no less than 70 percent in each of two written assessments that include multiple-answer questions and reflect the content of the aforementioned modules.

The link to the first assessment – that will be self-paced and non-proctored – will be sent by email on August 7 to eligible participants who will be given two weeks to submit their answers. Respondents will have the possibility to make two attempts to answer the assessment questions, and only the highest score will be considered.

Important note: For participants who wish to obtain a certificate of achievement, to be able to track your attendance, you must register and join the webinars using the same e-mail address.

Session 1

Speakers and moderator



Sarah Daniel

Is Research Assistant at the Climate Change and Natural Resource Sustainability Cluster, United Nations Economic and Social Commission for Western Asia (ESCWA). With a background in agricultural engineering and a master's degree in European forestry, Sarah has worked on planning and implementing several developmental projects in the field of agriculture and environment, in collaboration with relevant ministries in Lebanon and international organizations. During her work within ESCWA since 2020, she has been involved in researching, developing and delivering knowledge material to local and national agricultural stakeholders from the Arab region on the adoption of green technologies to enhance the resilience of the agricultural sector, in addition to researching other topics related to sustainable development such as food and water security, circular economy and sustainable consumption and production, among others.



Mohamed Abd Salam El Vilaly

Received a B.Sc. in Water Resources Management (University of Nouakchott, Mauritania, 2001), a Specialised M.Sc. degree in Groundwater Management (University of Grenoble, France, 2003), a Research M.Sc degree in Hydrology (University of Avignon and the Vaucluse, France, 2004) and a Ph.D. degree in Applied Remote Sensing for Earth Sciences (University of Arizona, United States, 2013). He has worked over 16 years in the humanitarian, academic, and private sectors. His research focuses on the integration and analysis of multitemporal airborne and satellite remote sensing data with GIS technologies to monitor and model human-environment interactions across a wide range of spatial and temporal scales. Specifically, Abd Salam's work intersects the concepts of big data analytics, decision making, planning and management, and the development of online data and knowledge exploration tools.



Maya Atie

Received her PhD in 2017 in Water Resources Engineering from the University of Guelph, Canada. She is the recipient of The Women in Science 2018 L'Oreal Fellowship and the Natural Sciences and Engineering Research Council of Canada Scholarship. Maya is also the founding president of the Lebanese Youth Parliament for Water advocating for sustainable management of water resources and youth inclusion. She is a part time lecturer and researcher at Notre Dame University and works as a consultant to different national and international organizations. Maya's work specializes in integrated water resource management and climate change and vulnerability assessments, as well as water and food security issues using a climate justice perspective.



Giulio Castelli

Is an Environmental Engineer, currently working on multiple research projects in East Africa, Central and South America. He holds a PhD in Sustainable Management of Agricultural Forestry and Food Resources, from the University of Florence (2018). During his PhD, he focused on water harvesting, water resources management and participatory approaches in international development. Alongside his PhD, he worked for the Italian Agency for Development Cooperation (AICS) in Sudan and as a consultant for the World Bank in Burundi. He is currently appointed as Research Fellow in Agricultural Hydraulics and Watershed Protection at the University of Florence Water Harvesting Lab.



Elena Bresci

Is Associate Professor at University of Florence in Italy (UNIFI). She graduated in Civil Engineering and got a Master's in Environmental Engineering in USA and PhD in Agricultural Engineering from UNIFI. She is the coordinator of the Water Harvesting Lab at UNIFI (<https://www.dagri.unifi.it/vp-261-wh-lab.html>). She teaches courses on water evaluation and management. Main research interests are land and water conservation measures under a vision of climate change, and water harvesting interventions.



Faraj El-Awar

Holds a PhD degree in water resources management. He started his career in academia where he joined the Department of Soil and Water Resources at the American University of Beirut. He developed a GIS-based hierarchical approach for selection of water harvesting reservoir sites in dry marginal lands of North Eastern Bekaa region of Lebanon. Dr. El-Awar moved to the UN system in 2001 as senior water expert in the humanitarian program for Iraq and later with UN-ESCWA and GIZ in Beirut, Lebanon. Dr. El-Awar established and headed the Global Water Operators' Partnerships Alliance (GWOPA) between 2018 and 2017. Dr. El-Awar is currently serving as Senior Water Specialist with FAO sub-regional office for Turkey and Central Asia.

Session 2

Speakers and moderator



Giulio Castelli

Is an Environmental Engineer, currently working on multiple research projects in East Africa, Central and South America. He holds a PhD in Sustainable Management of Agricultural Forestry and Food Resources, from the University of Florence (2018). During his PhD, he focused on water harvesting, water resources management and participatory approaches in international development. Alongside his PhD, he worked for the Italian Agency for Development Cooperation (AICS) in Sudan and as a consultant for the World Bank in Burundi. He is currently appointed as Research Fellow in Agricultural Hydraulics and Watershed Protection at the University of Florence Water Harvesting Lab.



Elena Bresci

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Theib Y. Oweis

Is a former director of the water, land, and ecosystems program of ICARDA and a world-leading specialist in water harvesting and management for agriculture, especially under conditions of water scarcity. He has a Ph.D. and MSc. degrees in Agriculture and Irrigation Engineering and spent over 35 years researching, developing, teaching, and publishing on water harvesting, supplemental irrigation, deficit irrigation, water productivity, salinity, and other areas associated with improving the productivity of water use in agriculture.



Waleed Hassan Abouelhassan

Is Senior Natural Resources Officer at FAO sub-regional Office for the Gulf Cooperation Council and Yemen. He is specialized in water resources management and holds two PhD degrees in water management from Egypt's Mansura University (2004) and Japan's Tottori University (2006). In cooperation with international water-related organizations, Waleed coordinated a number of international, regional, and national irrigation projects, including those involving integrated water resources, RS/GIS-based water project management, water reuse, water accounting, smart irrigation, water productivity, and solar-powered water pumping. Waleed has over 60 articles in international journals, papers, and conferences, and has supervised master's and doctoral theses.



Session 3

Speakers and moderator



Theib Y. Oweis

Is a former director of the water, land, and ecosystems program of ICARDA and a world-leading specialist in water harvesting and management for agriculture, especially under conditions of water scarcity. He has a Ph.D. and MSc. degrees in Agriculture and Irrigation Engineering and spent over 35 years researching, developing, teaching, and publishing on water harvesting, supplemental irrigation, deficit irrigation, water productivity, salinity, and other areas associated with improving the productivity of water use in agriculture.



Salim Roukoz

Has a M.Sc. in Water and Environmental engineering from Lebanese University and a diploma of Agricultural Engineering in irrigation from the Faculty of Agriculture. He is currently working as a GIS and irrigation expert at FAO Lebanon in a five-year forestry project (SALMA) "Smart Adaptation of Forest Landscapes in Mountain Areas" funded by GEF. Salim Roukoz has been the head of "Irrigation and Rural Engineering Service" at the Lebanese Ministry of Agriculture from 2006 till 2015. He has been conducting water and forestry projects and accumulating experience in the fields of GIS, irrigation, modelling, and climate change. Salim Roukoz has been involved, as an expert and national coordinator, in several international projects with EU, FAO, IFAD, USAID, GIZ, among others.



Elie Choueiri

Is currently Programme Associate at FAO Lebanon. With an Agricultural Engineer Degree and currently Master of Science in Management and Food Quality Control, he is in charge of the follow-up of the project's activities on "Implementing the 2030 Agenda for water efficiency/productivity and water sustainability in NENA countries" with its both components water accounting/water auditing and water productivity. He is also the Food Security Sector Co-Coordinator under the Lebanon Crisis Response Plan of the Syria Crisis (FAO Co-lead). In addition, he is in charge of following the different projects implemented by FAO Lebanon in support to the Ministry of Agriculture in terms of carrying out agriculture census, agricultural production surveys as well as the farmers' households surveys to better understand the impact of current multiple shocks Lebanon is facing on their Food Security situation and agricultural livelihoods.



Module 4: Main Guidelines for Planning and Designing RWH Systems

M4/S1: Technical, Socio-economic and Environmental Considerations in Planning and Designing RWH Systems

Date: 19th July, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	Welcoming Remarks Introduction to the session objectives and speakers Moderator: Faraj El Awar, Senior Land and Water Specialist, FAO-SEC
01:05 – 01:30	Presentation on “Planning and design of rainwater harvesting systems for Enhancing Resilience and Sustainability of Agriculture in the Arab region: socio-economic and environmental considerations” By: Sarah Daniel, Research Assistant at the Climate Change and Natural Resource Sustainability Cluster, ESCWA Maya Ati, Consultant in Integrated water Resource Management and Climate Mohamed Abd Salam El Vilaly, Water Resource Management Analyst, ESCWA
01:30 – 01:55	Presentation on “Participatory Approach to Water Harvesting implementation” By: Giulio Castelli and Elena Bresci, Università degli Studi di Firenze
01:55 – 02:25	Open Discussions – Q&A's
02:25 – 02:30	Conclusion & Wrap-up



Module 4: Main Guidelines for Planning and Designing RWH Systems

M4/S2: Planning and Designing of RWH Systems and Structures

Date: 26th July, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	Welcoming Remarks Introduction to the session objectives and speakers Moderator: Waleed Hassan Abouelhassan, Senior Natural Resources Officer at FAO sub-regional Office for the Gulf Cooperation Council and Yemen
01:05 – 01:30	Presentation on “Best Siting of Water Harvesting Structures” By: Giulio Castelli and Elena Bresci, Università degli Studi di Firenze
01:30 – 01:55	Presentation on “Planning and Designing in-situ Rainwater Harvesting Systems for Agriculture” By: Theib Oweis, ex-ICARDA, Egypt
01:55 – 02:25	Open Discussions – Q&A's
02:25-02:30	Conclusion & Wrap-up



Module 4: Main Guidelines for Planning and Designing RWH Systems

M4/S3: Planning and Designing of RWH Systems

Date: 2nd August, 2022

Time: 01:00 PM - 02:30 PM (GMT +2)

Time	Activity
01:00 – 01:05	Welcoming Remarks Introduction to the session objectives and speakers Moderator: Elie Choueiri, Programme Associate at FAO Lebanon
01:05 – 01:30	Presentation on “Planning and Designing Vallerani Systems in Jordan - Lessons Learnt” By: Theib Oweis, ex-ICARDA, Egypt
01:30 – 01:55	Presentation on “Water harvesting for forest fire prevention in Lebanon” By: Salim Roukoz, GIS and Irrigation expert at FAO Lebanon, FAO Lebanon
01:55 – 02:25	Open Discussions – Q&A's
02:25-02:30	Conclusion & Wrap-up



This activity is implemented within the regional project "Implementing the 2030 Agenda for water efficiency/productivity and water sustainability in NENA countries" under the umbrella of the Water Scarcity Initiative. This regional project is implemented by the Food and Agriculture Organization of the United Nations and funded by the Swedish International Development Cooperation Agency (SIDA).

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